USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 14 LAB - HD RADIO
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 6.3: I can describe how HD Radio works and demodulate an HD Radio signal.

This packet is the L14 Lab, in three parts, and you turn it in once, complete, before Lesson 15. Do **Part 1** on your own before the lab period: run the sideband finder and make the predictions you will use at the bench. **Part 2** is your own measurements, recorded in class on your own laptop with your RTL-SDR. Do **Part 3** on your own after the lab: compare the measurements with the rules and answer the analysis questions.

Grading. One submission, graded against Objective 6.3 on the course's 1 / 0 scale, with revision allowed under course policy. Paste screenshots where a question asks for one.

Transmitting. Nothing in this lab transmits. The RTL-SDR receives only, and both captures came from GNU Radio flowgraphs, not from an antenna.

The flowgraphs are on the L14 lesson page, `L14-hd-test-tx.grc` and `L14-hd-find.grc`, and the capture `L14-lab-capture.cu8` is on the L14 Lab page. Both captures are simulated: unsigned 8-bit I and Q at 1,488,375 samples per second, the format the RTL-SDR and nrsc5 record. Every number below uses the hybrid values from the lesson: subcarrier spacing $\Delta f = 1{,}488{,}375/4096 = 363.37$ Hz, sidebands from subcarrier 356 to 546 on each side, partitions of 19 subcarriers, and an extended hybrid station adding one, two, or four partitions inside subcarrier 356 on each side.

Part 1: Pre-Lab (individual, before the lab period)

1. **Run the sideband finder.** Put both L14 flowgraphs in one folder. Run `L14-hd-test-tx.grc` once, then open and run `L14-hd-find.grc`.

   (a) After about 5 s, record the two meter readings.

   upper [ ]

   lower [ ]

   (b) Paste a screenshot of the running flowgraph's window.

2. Predictions. You will use each of these at the bench.

(a) A station adds one, two, or four partitions inside subcarrier 356 on each side. For each, find the new inner subcarrier number, its frequency from the channel center, and the number of subcarriers in one sideband.

Added partitions	1	2	4
Inner subcarrier			
Inner edge (kHz)			
Subcarriers per side			

(b) Find the frequency sink's bin width, the number of bins one 69,041 Hz sideband spans, and the number of subcarriers per bin, for an FFT of 1024 points and of 8192 points at 1,488,375 samples per second.

FFT size	Bin width (Hz)	Bins per sideband	Subcarriers per bin
1024			
8192			

(c) A hybrid station runs its digital signal at -14 dBc, the level the FCC allows by notification, with equal sidebands. Find each sideband's level and each subcarrier's level in dBc. If its analog signal is 6 kW, how many watts is the digital signal in total?

sideband

subcarrier

total

(d) An extended hybrid sideband with two added partitions has the same power in every subcarrier. A meter set to subcarriers 356 to 546 measures only part of it. By how many dB does that meter read low?

Part 2: Measurements (in class)

3. **Build the sideband meter.** Start GRC with an empty canvas. Do not open `L14-hd-find.grc`: build the meter yourself as the lab page lists.

   (a) Paste a screenshot of your flowgraph on the GRC canvas. Every connection must be black, not red.

   (b) Run it on `L14-hd-test.cu8` and record both readings. Paste a screenshot of the running window.

   upper

   lower

4. Identify the lab capture. Run your meter on `L14-lab-capture.cu8`, as the lab page lists. Treat the center as 100.3 MHz.

(a) Zoom in on each sideband's edges with an FFT of 8192 points, read them, convert each to a subcarrier number, and name the configuration.

	Lower outer	Lower inner	Upper inner	Upper outer
Edge (MHz)				
Subcarrier				

(b) Record both meter readings with `inner` at 356 and with `inner` set to the inner edge you found.

`inner`	Upper (dBc)	Lower (dBc)
356		
your inner edge		

(c) Find the total digital power in dBc, and the difference between the sidebands.

total

difference

(d) Does the station 400 kHz above the center carry HD Radio? How can you tell?

5. A live station. Add the RTL-SDR Source to your meter and find a local station with HD Radio, as the lab page lists.

(a) Paste a screenshot with the shelves visible, and record the station, the RF gain, and the height of the shelves above the floor.

(b) Record the inner and outer edges, the subcarrier numbers, and the configuration.

(c) Record both meter readings and the total digital power in dBc.

upper	
lower	
total	

6. Decode it. Decode the same station with nrsc5, as the lab page lists.

(a) Record the station name, the title and artist of what is playing, and the programs the station carries.

(b) Lower the gain until the audio stops. At what gain does it stop, and how high do the shelves stand above the floor at that gain in your meter?

(c) Record 10 s with `nrsc5 -w`, open the recording in your meter, and record both readings. Do they match Question 5(c)?

upper

lower

Part 3: Writeup (individual, after the lab)

7. Analysis.

(a) Compare the live station's sideband levels with the original -20 dBc, the -14 and -10 dBc levels the FCC allowed in 2010, and the NRSC-5 limit of 10 dB on asymmetry. How is the station configured?

(b) Explain how the lab capture's edges and meter readings identified its configuration, and give one reason a station would choose each feature you found.

(c) nrsc5 cannot decode `L14-hd-test.cu8`, although its sidebands have the right subcarriers, spacing, and guard. Name two things nrsc5 needs from a signal that the test file does not provide.

Documentation: _____